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Skew-step function governing main and intermediate intensity spots in radial carpet beams

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Abstract

Radial carpet beams (RCBs) are produced by the diffraction of a plane wave from radial structures, resulting in unique properties such as self-healing, non-diffracting behavior, accelerating propagation, core-area amplifying, and discrete intensity patterns. While the mathematical formulation of RCBs is well-established, the precise behavior of their intermediate radial intensity spots as the number of grating spokes varies remains underexplored. In this study, we investigate the relationship between the number of grating spokes and the generated spots in the intermediate radial distances. Interestingly, the number of intermediate radial intensity peaks increases with the number of grating spokes, not in a smooth linear fashion but through a series of tilted steps, each slightly lower than the end of the previous one, showing a peculiar yet structured behavior. We reveal a pair of skew-step functions that govern this dependency, with both functions having the same form but being laterally shifted. This demonstrates an inherent order similar to systematic natural phenomena, such as Fibonacci-based phyllotaxis in plants, spiral shell formations, planetary spacing described by the Titius–Bode law, and digit distributions predicted by Benford’s law. This work provides new insights into the structured behavior of RCBs, enhancing our understanding of the underlying principles governing optical beam dynamics and their parallels to natural systems.

Supplementary material for this article is available [online](#)

Keywords: structured light, radial carpet beams, discrete optical patterns, skew-step function, beam shaping, hidden regularities in optical beams

1. Introduction

Structured beams are optical fields characterized by complex distributions of intensity, phase, and/or polarization [1, 2]. Many of these beams exhibit remarkable properties such as non-diffraction, self-healing, and self-acceleration. For

example, Bessel, vortex, and Airy beams maintain their shapes during propagation, allowing them to reconstruct obstructed regions and resist amplitude and phase distortions [3]. These features make them ideal for imaging through turbid media, with notable applications in biological microscopy, where Bessel beams enhance imaging depth and resolution [4, 5]. Recently, the superposition of orthogonally polarized Bessel modes has enabled the generation of optical skyrmions with similar non-diffracting and self-healing properties

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[6]. Self-accelerating beams, such as Airy beams, follow curved parabolic trajectories in free space, enabling applications like optical trapping, where particles are guided along predefined curved paths [3, 5]. Certain structured beams, including Hermite–Gaussian (HG), Laguerre–Gaussian (LG), and higher-order Airy and Bessel beams, exhibit multiple intensity spots or rings in their transverse profiles, enabling diverse applications such as information transmission and multi-particle trapping. In HG and LG beams, the number of these spots is determined by clear mathematical relationships [7–9]. For instance, HG beams allow simultaneous trapping of multiple particles [10, 11], while LG beams, owing to their orbital angular momentum, are employed in free-space optical communication for data encoding and in optical tweezers for particle rotation [5, 12, 13]. Higher-order Bessel and circular Airy beams, with their intrinsic ring-shaped intensity distributions, are particularly advantageous for trapping applications [14, 15].

Among these structured beams, the radial carpet beams (RCBs) stand out due to their unique and intricate cylindrical symmetry, featuring numerous intensity spots [16]. Despite exhibiting complex intensity and phase patterns, their generation does not require sophisticated optical setups; they can be produced simply by the diffraction of a plane wave from a radial amplitude/phase grating [16, 17]. Notably, this family of structured beams simultaneously exhibits non-diffracting, self-healing, and self-accelerating behaviors. Remarkably, in addition to these properties, they also demonstrate local self-amplifying, making them particularly unique and compelling [16–19]. They are less diffracting due to the fact that, although the beam size increases during propagation, its rate of expansion is slower than that of a Gaussian beam and gradually decreases with distance [16, 18]. By applying the stationary phase principle, the beam can even be engineered to exhibit zero divergence [20], further emphasizing its non-diffracting nature. Exhibiting accelerating behavior, each point within the intensity profile follows a curved trajectory during propagation, appearing to be drawn inward toward the optical axis [16]. These beams possess remarkable self-healing capabilities, such that both theoretical analyses and experimental results have shown they can reconstruct their structure even after significant perturbations, such as blocking a main intensity spot (MIS) or multiple regions of the intensity pattern [16, 19]. Their classification as locally self-amplifying beams arises from the observation that the power in their core area increases due to the redistribution of intensity from the empty central region [21]. As a clarifying explanation, while the MISs expand during propagation, their maximum intensity remains constant. Consequently, the total power confined in each spot grows with distance as energy is continuously transferred from the patternless central region into the structured areas of the beam [21]. Additionally, RCBs demonstrate superior resilience to turbulence compared to other structured beams, particularly due to their complex structure, which results in less than 1% (almost zero) variation in MISs even under high turbulence, unlike LG beams, which experience significant

deformation [22, 23]. Recently, new families of RCBs have been proposed, including elliptical RCBs [24] and fractional RCBs [25], which extend the versatility and application scope of this beam class.

One of the most distinctive features of RCBs is the discrete arrangement of intensity spots along the radial direction, where the number of spots increases incrementally as m , the number of spokes in the radial grating, grows. RCBs generated from a binary radial phase grating with m spokes produce $2m$ azimuthal intensity spots, symmetrically distributed on concentric circles within the middle region of the beam's intensity profile, with the number of these circles represented by n . At the outer boundary of that area, a ring of $4m$ azimuthal spots forms. However, the precise mathematical relationship between n and m is not fully understood, with preliminary findings indicating non-trivial non-linear behavior. Understanding these underlying mathematical relationships is key to deepening our comprehension of RCBs and their structured nature. This study investigates this relationship in detail, offering deeper insights into the structured patterns formed by RCBs.

Similar mathematical investigations in natural systems, such as the Fibonacci sequence and the golden ratio, have provided significant insights into natural periodicity and symmetry, demonstrating the inherent mathematical order present in biological growth patterns like the arrangement of leaves around a stem (phyllotaxis), the patterns of seeds in sunflowers, and the spiral shells of snails [26–29]. Surprisingly, the Fibonacci sequence also appears in the reflection of light rays between two plates of glass, where the number of rays emerging corresponds to Fibonacci numbers depending on the number of allowed reflections [26]. Likewise, the Titius–Bode law, which describes the approximate regularity in the sequence of distances from the Sun of the planets in our Solar System, gained credibility with the discoveries of Uranus and Ceres, and aided in the prediction of Neptune [30, 31]. In some cases, new mathematical relationships have been discovered unexpectedly, such as the accidental derivation of a novel formula for π by string theorists [32, 33]. Additionally, Benford's Law, which predicts the distribution of first digits in naturally occurring datasets, demonstrates that even random data often follows predictable mathematical patterns, hinting at an underlying order in nature's numbers [34, 35].

Such patterns are found in other discrete and periodic systems, such as the Lotka–Volterra equations, which not only describe regular oscillations in predator-prey dynamics, but are also widely used to model the dynamics of ecological communities by accounting for species interactions and migration [36–38]. Recent research proposing a universal biological law to explain aging also demonstrates how fundamental mathematical relationships govern various complex natural systems [39, 40].

In this work, we aim to uncover the mathematical relationship between n and m in RCBs. Our investigation reveals that as the number of spokes in the radial binary phase grating increases, the number of intermediate radial intensity spots

in the generated beam follows a systematic and discrete pattern. These mathematical rules governing the beam's intensity distribution emphasize the inherent order and structure in RCBs, contributing to a deeper understanding of their nature and underlying regularity. By uncovering these mathematical connections, this research clarifies the structured behavior of RCBs and lays the groundwork for practical applications, such as beam shaping, optical communication, multi-particle trapping, and high-resolution imaging. The latter is made possible by their multiple localized high-intensity spots that are symmetrically distributed in polar coordinates. This spatial configuration resembles structured beam patterns used in structured illumination microscopy (SIM) systems, where the illumination is translated across the field of view to access high-frequency sample information more effectively [41, 42]. For example, one study used a digitally shifted multifocal lattice generated by a digital micrometer device, where the pattern was translated pixel by pixel under digital control to scan the sample [41]. Another study employed a three-beam interference pattern, in which the period and orientation were adjusted using galvanometer mirrors and the phase was shifted via piezo stages [42]. Although RCBs exhibit a central patternless region with zero intensity, their structured spatial profile makes them suitable for SIM. Inspired by these strategies, RCBs offer a similarly structured yet flexible form of illumination. By employing radial gratings with a small number of spokes, the radial spacing between adjacent intensity spots can be reduced, enabling finer radial sampling of the sample. The central point on the optical axis can be illuminated by using the beam without a radial grating, producing a point-like illumination. Furthermore, by sequentially illuminating the sample with RCBs generated from gratings of different spokes (m), and rotating the radial grating in its plane, both radial and angular positions can be scanned. This approach, which requires only digital updates to the spatial light modulator (SLM) and no moving parts, ensures comprehensive field coverage and enables high-resolution imaging across the entire sample. The significance of this research lies in uncovering the mathematical rules that govern the intensity distributions of RCBs, highlighting the systematic and discrete nature of these patterns.

2. Diffraction from radial structures

Here, we present a concise overview of the diffraction theory for radial structures to introduce the complex amplitude and the principles behind RCB generation. Radial structures are periodic formations consisting of radial lines that exhibit angular periodicity, where the transmission function is independent of the radial distance [18, 21]. Considering (r, θ) as polar coordinates, the transmission function of a radial structure can be represented as [16, 18]

$$t(\theta) = t_0 + \sum_{s=1}^{+\infty} (t_s e^{is\theta} + t_{-s} e^{-is\theta}), \quad (1)$$

where t_s represents the Fourier coefficient of order s .

When illuminated by a plane coherent wave of wavelength λ , the complex amplitude of the diffracted light at a distance z is computed using the Fresnel–Kirchhoff diffraction integral [18]:

$$U(r, \theta, z) = e^{ikz} \left\{ t_0 + \sqrt{\rho} e^{i\rho} \sum_{s=1}^{+\infty} \sqrt{\frac{\pi}{2}} (-i)^{\frac{s}{2}+1} \times (t_s e^{is\theta} + t_{-s} e^{-is\theta}) \left[J_{\frac{s+1}{2}}(\rho) + i J_{\frac{s-1}{2}}(\rho) \right] \right\}, \quad (2)$$

where $\rho = \frac{kr^2}{4z}$ is a dimensionless parameter [18], $k = \frac{2\pi}{\lambda}$ is the wave number, and $J_{\frac{s\pm 1}{2}}(\rho)$ is the Bessel function of the first kind of order $\frac{s\pm 1}{2}$.

A radial binary phase structure consisting of m spokes can be described as [16]

$$t(\theta) = \exp(i\gamma \operatorname{sgn}[\cos(m\theta)]), \quad (3)$$

where γ is the phase modulation amplitude, and sgn represents the sign function. The phase distribution of the radial binary phase grating is illustrated in the figure 1(a) for $m = 25$ and $\gamma = \frac{\pi}{2}$.

The Fourier series expansion for the transmission function equation (3) can be expressed as [16]

$$t(\theta) = \cos(\gamma) + \sum_{\substack{l=1 \\ \text{odd}}}^{\infty} \frac{2}{l\pi} i^l \sin(\gamma) (e^{ilm\theta} + e^{-ilm\theta}). \quad (4)$$

By comparing the equation (4) with the equation (1), it can be inferred that $t_0 = \cos(\gamma)$, $t_{s=lm} = t_{s=-lm} = \frac{2}{l\pi} i^l \sin(\gamma)$, and all other coefficients are zero. Thus, substituting these coefficients into equation (2), the complex amplitude of the beam produced by the diffraction of a plane wave from the structure given by equation (3) can be described by [16]

$$U(r, \theta, z) = e^{ikz} \left\{ \cos(\gamma) + \sqrt{\rho} e^{i\rho} \sum_{\substack{l=1 \\ \text{odd}}}^{+\infty} u_l \times \left[J_{\frac{lm+1}{2}}(\rho) + i J_{\frac{lm-1}{2}}(\rho) \right] \cos(lm\theta) \right\}, \quad (5)$$

where $u_l = \frac{2}{l} \sqrt{\frac{2}{\pi}} \sin(\gamma) (-i)^{(\frac{m}{2}-1)l+1}$. While infinite plane waves and gratings are assumed in theory, practical implementations with finite beams limit the propagation distance over which RCBs preserve their structure. This distance is approximately $z = \frac{2\pi^2 r_0^2}{m^2 \lambda}$, where r_0 is the plane wave radius [21]. For example, with $r_0 = 3$ cm and $\lambda = 532$ nm, the structure is maintained up to about 83 m for $m = 20$ and around 9 m for $m = 60$. It is worth noting that illuminating a radial grating with a Gaussian beam produces a petal-like diffraction pattern in the near-field regime, which eventually transforms into a Gaussian beam in the far field [43]. Therefore, while finite plane wave illumination of a radial grating generates an RCB that preserves its structure over a finite propagation distance, Gaussian illumination results in a fundamentally different behavior.

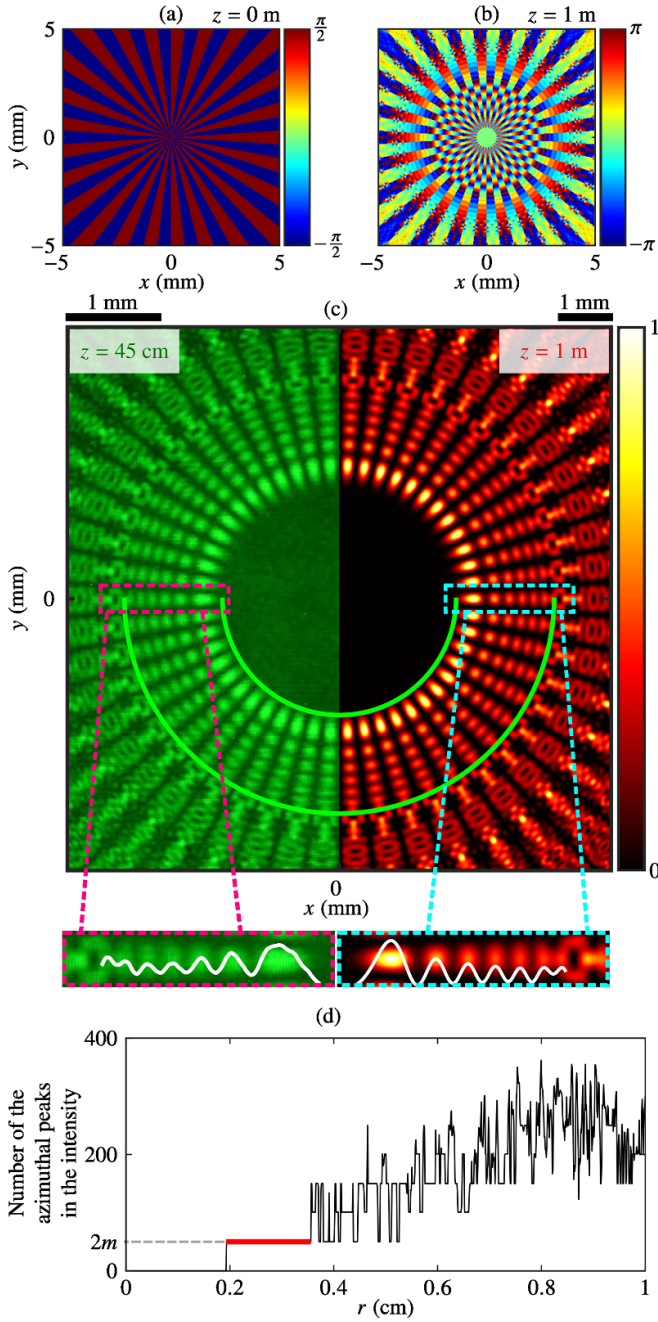


Figure 1. (a) Phase distribution of the radial binary phase grating with $m = 25$ and $\gamma = \frac{\pi}{2}$. (b) Theoretically plotted phase pattern of the generated beam after diffraction by the grating in (a) at $z = 1$ m. (c) Experimental (green, left) intensity pattern of the generated beam at $z = 45$ cm and the corresponding theoretical (red, right) pattern at $z = 1$ m, illustrating the excellent agreement between theory and experiment. The propagation distances z for the experimental and theoretical patterns were not chosen to be identical due to experimental limitations on the available propagation range. The magnified sections below (c) provide a close-up view of radial intensity variations to highlight intermediate spot details. See also Visualization 1 for an illustration of the beam propagation up to 1 m (Multimedia available online). (d) Number of azimuthal intensity peaks versus radial distance for the cross-sectional intensity profile of the beam with $m = 25$ and $\gamma = \frac{\pi}{2}$ at $z = 1$ m, showing the specific radial range where $2m$ intensity spots are consistently observed.

The theoretically calculated phase and intensity profiles of this beam having a wavelength of 532 nm, after propagation of 1 m from the grating defined by equation (3) with $m = 25$ and $\gamma = \frac{\pi}{2}$, are depicted in figures 1(b) and red part of 1(c), respectively. In figure 1(c), the green part (left) illustrates the experimentally recorded intensity pattern of the same beam at a distance $z = 45$ cm from the same grating. The experimental setup includes passing the diode-pumped laser beam of wavelength 532 nm (Nd:YAG) through a spatial filter, collimating it using a lens, and diffracting it through different radial binary phase gratings displayed on an SLM (3 M X50, resolution: 1024×768 , display: 0.7 in, polysilicon LCD). The resulting pattern is captured by a high-resolution camera (Nikon D7200). As can be seen, the scale bars in the two parts of figure 1(c) differ because the experimental window is smaller. This is due to the limited dimension of the used SLM as will be discussed in detail in section 4. Here, to preserve the beam's structure, we experimentally recorded the RCB patterns at $z = 45$ cm from the grating, in which the observed window is smaller than the theoretically calculated patterns. Importantly, the beam's structural preservation during propagation can still be inferred even from the comparison between the theoretical pattern at $z = 1$ m and the experimental pattern at $z = 45$ cm. Magnified sections below figure 1(c) provide close-up views of radial intensity variations, highlighting intermediate spot details. Furthermore, Visualization 1 illustrates the beam's propagation up to a distance of 1 m, demonstrating the preservation of its transverse intensity profile during propagation (Multimedia available online).

3. Numerical determination of the radial range with $2m$ intensity spots

The beam described by equation (5) generates an intensity pattern characterized by an array of $2m$ azimuthal intensity spots arranged on concentric circles within a specific radial range. The enlarged patterns in figure 2, derived from figure 1(c), provide a detailed view of this structure. In figure 2(a), the MISs, which predominantly exhibit the maximum intensity across the cross-sectional pattern, are enclosed by cyan semicircles. The region containing $4m$ azimuthal spots is highlighted by green semicircles. Additionally, figure 2(b) shows concentric dashed cyan semicircles, each comprising $2m$ azimuthal intensity spots, and highlights the intermediate radial distance $[r_1, r_2]$ using solid green semicircles.

To precisely determine this radial range, a MATLAB script was developed to analyze the beam's intensity distribution in detail. Specifically, the code calculates the azimuthal intensity profile at each radial distance, then detects and counts the number of intensity peaks for each profile. The goal was to identify the radial interval $[r_1, r_2]$ where $2m$ azimuthal intensity spots are consistently present, representing a significant feature of the diffraction pattern.

During this analysis, we set the phase modulation value to $\gamma = \frac{\pi}{2}$ to simplify the mathematical model and ensure

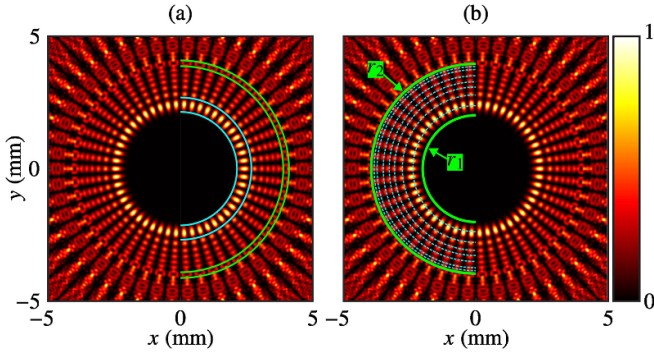


Figure 2. Intensity pattern analysis for RCB with $m = 25$ and $\gamma = \frac{\pi}{2}$ at $z = 1$ m. (a) The cyan semicircles indicate the region containing the $2m$ MISs, while the green semicircles enclose the region with $4m$ azimuthal intensity spots. (b) The cyan dashed semicircles depict seven concentric regions, each containing $2m$ azimuthal intensity spots, while the green solid semicircles mark the intermediate radial range $[r_1, r_2]$, defining the radial interval where the $2m$ azimuthal spot arrangement is consistently observed.

consistent results across different values of m . An example output of this analysis is illustrated in figure 1(d) which shows the number of azimuthal intensity peaks, as a function of radial distance for the beam with $m = 25$ at a propagation distance of $z = 1$ m. In this graph, the consistent occurrence of $2m$ azimuthal peaks over a specific radial range is clearly highlighted.

The corresponding cross-sectional intensity profile of the beam is shown in figure 1(c), where concentric rings with $2m$ intensity spots are bounded by two green semicircles. To determine the exact boundaries of this region, we used MATLAB to analyze radii ranging from 0 to 1 cm. The horizontal red segment in figure 1(d) indicates the identified radial range, denoted as $[r_1, r_2]$, where the $2m$ azimuthal peaks are consistently present. This range allows us to define the spatial extent of this unique intensity distribution pattern.

Once the radial range was identified, the next step was to calculate the number of intermediate radial intensity spots, denoted by n . This calculation was based on the number of maxima detected in the radial range shown in figure 1(d). The process was repeated for different values of m , ranging from 2 to 120, to investigate how the radial range and intensity patterns vary with different beam parameters.

The results of these calculations and their experimental validation are presented in figure 3, where the cross-sectional intensity profiles of several RCBs are shown. The top two rows correspond to theoretically plotted beams, while the bottom two rows show the experimentally generated beams. For each beam, the specific radial range $[r_1, r_2]$ is marked by two green circles in the theoretically plotted beams and pink circles in the experimentally generated beams, clearly illustrating how the region containing $2m$ azimuthal intensity spots varies with different values of m . For each case, a dashed rectangle highlights a selected area of the beam, and the corresponding enlarged view is shown directly below as an inset. The intensity profile along a transverse line passing through the central row of intermediate radial spots is overlaid as a solid white curve in the insets, shown for both the theoretical and experimental cases.

For the experimental intensity plots, it is important to note that these intensity profiles have been smoothed to suppress fluctuations caused by noise in the recorded images. Comparing the experimental and theoretical insets and the corresponding one-dimensional intensity profiles, we observe that the number of peaks in the intensity plots, and thus the number of radial intermediate spots (n) within the selected radial interval ($[r_1, r_2]$) matches perfectly, further validating the consistency between theory and experiment.

This representation highlights the relationship between the number of spokes in the radial phase grating and the resulting intensity pattern, providing insight into the underlying diffraction behavior. Additionally, Visualization 2 provides a video sequence of the beam intensity patterns for radial binary phase gratings with spoke numbers ranging from $m = 2$ to 120, including both the theoretically plotted patterns at a propagation distance of $z = 1$ m and the experimentally generated patterns at $z = 45$ cm (Multimedia available online). As previously explained, due to the limited dimensions of the SLM used, experimental imaging for all values of $m = 2$ to 120 was performed at $z = 45$ cm to preserve the beam's structure. Consequently, the experimental window is smaller than the theoretical computational window.

It is important to note that the complex amplitude of the beams under study, as described by equation (5), is intricately dependent on the parameters z , r , and λ through the dimensionless parameter $\rho = \frac{kr^2}{4z}$, with z and λ serving as scaling factors for the beam profile. This relationship implies that the overall shape of the beam is preserved under propagation, with changes in z and λ affecting only its scale rather than its internal structure [16, 44]. To further illustrate this property, figure 4 shows the intensity patterns of the beam for $m = 10$ and $\gamma = \pi/2$ under different conditions. The first row displays patterns at various propagation distances with a fixed wavelength of $\lambda = 650$ nm, while the second row presents patterns for different wavelengths at a fixed propagation distance of $z = 1$ m. These results clearly demonstrate that the beam's structural features remain unchanged, with both the wavelength and propagation distance acting as scaling parameters.

4. Variation trends in the number of intermediate radial intensity peaks

Next, we analyze how the number of intermediate radial intensity peaks, n , varies with the beam parameter m . Using MATLAB, we computed n within the radial range where $2m$ azimuthal intensity spots consistently appear for each m , and the resulting graph, $n(m)$, is shown in figure 5.

Although good agreement between the theoretical and experimental intensity patterns is evident in figures 1 and 3, it should be noted that the SLM used to generate the beams had dimensions of $1.2 \text{ cm} \times 1.6 \text{ cm}$, and its diffraction orders could be isolated from each other at approximately $z = 45$ cm. By implementing the radial phase grating on the SLM, an RCB is generated in each of these diffraction orders. Due to the limited size of the SLM (1.2 cm), and because the

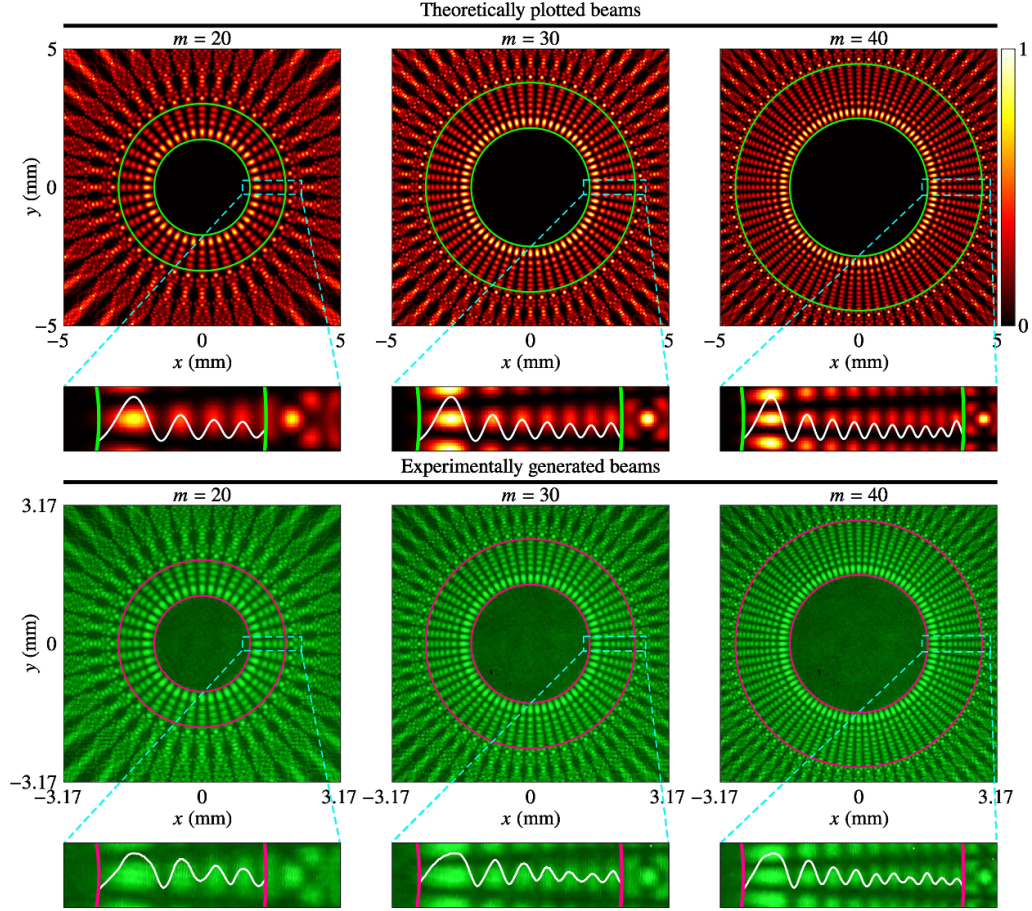


Figure 3. Cross-sectional intensity profiles of RCBs for various values of m ($m = 20, 30, 40$) and $\gamma = \pi/2$. The top two rows show theoretically plotted beam patterns at $z = 1$ m, with green circles indicating the radial range $[r_1, r_2]$, where an array of $2m$ azimuthal intensity spots is consistently observed on concentric rings. The bottom two rows display the corresponding experimentally generated beams at $z = 45$ cm, illustrating close agreement with the theoretical predictions. Insets in each case represent the enlarged view of the dashed rectangle shown in the corresponding overall beam intensity pattern, with white solid lines representing the intensity profile along a transverse cut passing through the central row of spots. For the experimental plots, these profiles have been smoothed to reduce noise-related fluctuations in the recorded data. See also Visualization 2 for a video sequence of beam intensity patterns at $z = 1$ m, covering spoke numbers from $m = 2$ to 120 (Multimedia available online).

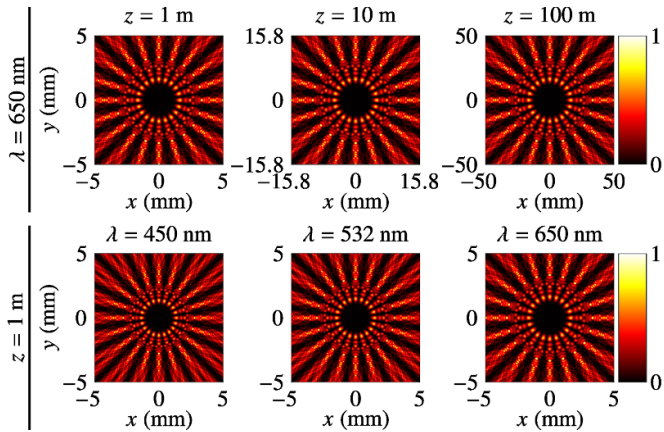


Figure 4. Intensity patterns of the RCB for $m = 10$ and $\gamma = \pi/2$, illustrating propagation distance effects (top row: $z = 1, 10, 100$ m at $\lambda = 650$ nm) and wavelength effects (bottom row: $\lambda = 450, 532, 650$ nm at $z = 1$ m).

RCB expands during propagation, causing r_1 , r_2 , and $r_2 - r_1$ to increase, the outer radius r_2 for experimentally generated beams with $m > 70$ at $z = 45$ cm extends beyond the diffraction window defined by the illuminated area on the SLM. As the beam propagates, its transverse size increases; however, the similarity of its intensity profile to the ideal pattern remains unchanged. Thus, within a window equivalent to the initial illuminated aperture on the SLM (the diffraction window), only the central portion of the beam, which has expanded to match the size of this window at each propagation distance, preserves the characteristics of the original pattern. As a result, the diffraction pattern of an experimentally generated RCB matches its ideal profile only within the diffraction window [21]. In other words, to produce an RCB with an ideal profile over a larger transverse area and allow it to propagate farther without losing this profile, it is necessary to illuminate a larger radial grating with a wider plane wave that fully covers the grating. This limitation and its effect on the experimental

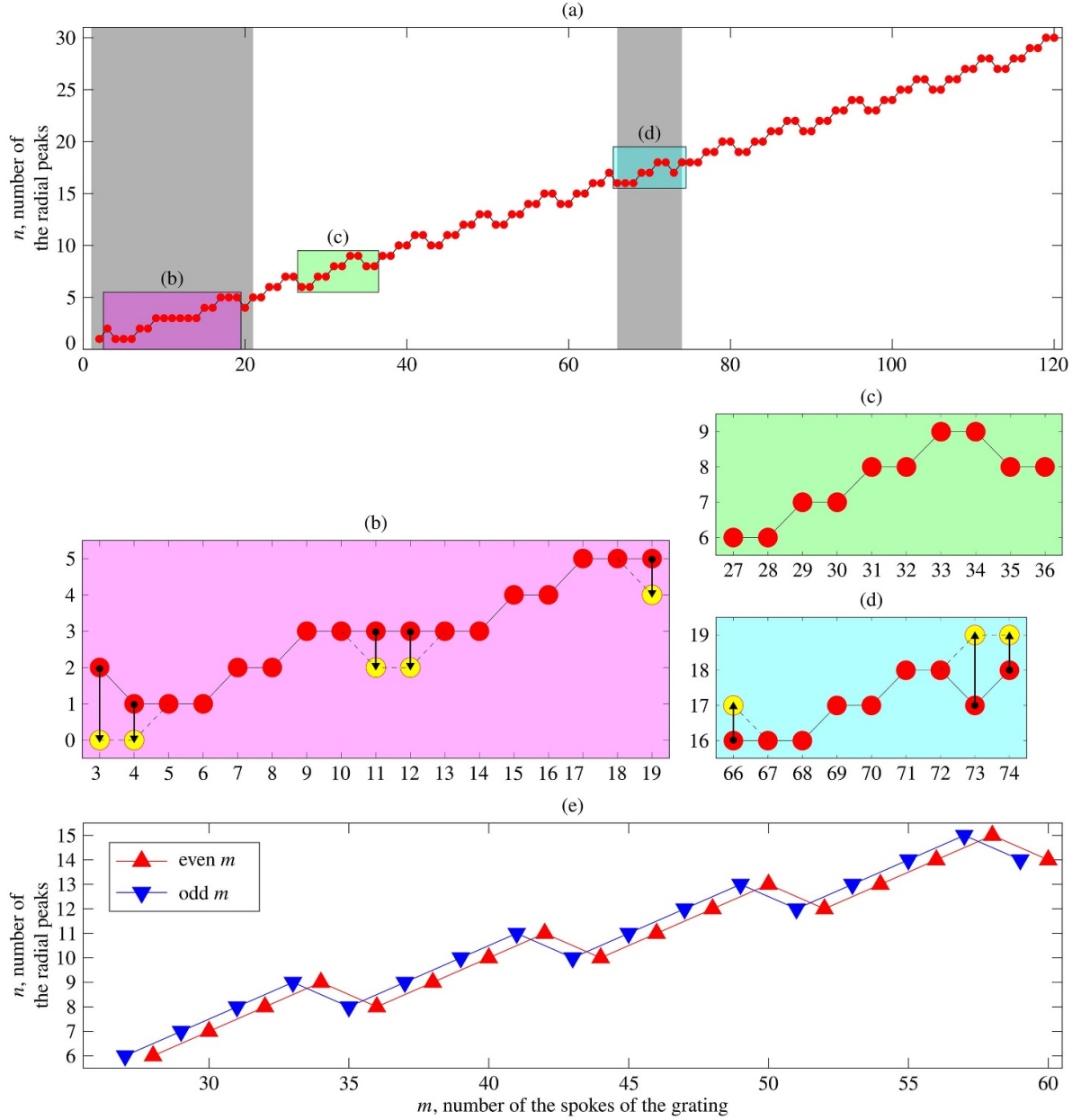


Figure 5. (a) Number of intermediate radial peaks, n , as a function of the number of grating spokes, m , showing the overall trend (red circles) and highlighting irregular regions (shaded gray). Insets provide magnified details: (b) pink shaded area ($m = 3$ – 19) shows fluctuations around initial values, with yellow circles marking expected positions if the trend had been followed; (c) green shaded area ($m = 27$ – 36) shows a close-up of a periodic segment; (d) blue shaded area ($m = 66$ – 74) highlights deviations from the expected trend, with yellow circles indicating where points should align for continuity. (e) Bottom panel shows a separate analysis of even (red triangles) and odd (blue triangles) m values, illustrating the periodic nature of peak variation. This analysis reveals a non-trivial relationship between intermediate and main intensity spots in RCBs, characterized by a pair of laterally shifted skew-step functions.

results are evident in Visualization 2, where the experimental window becomes progressively smaller and incomplete as m increases.

In addition, we attempted to perform computational analysis in MATLAB on the experimentally recorded beam images. This included accurate center detection, background subtraction based on the average intensity of the central patternless region, identification of the radial interval $[r_1, r_2]$, and finally, counting the number of intermediate radial intensity spots (n). Despite extensive efforts to apply smoothing and

noise reduction while preserving structural details and avoiding artificial blurring, the analysis faced several challenges. The limited size of the SLM, and consequently the low resolution of the applied radial phase pattern, leads to the generated beam deviating from the ideal theoretical profile (which is rendered with $2^{11} \times 2^{11}$ resolution in the theoretical beam calculation). Even if this issue is mitigated, the limited resolution and presence of dead pixels in the camera introduce further degradation in image quality. Additionally, local camera vibrations during capture sometimes cause elongation or blur

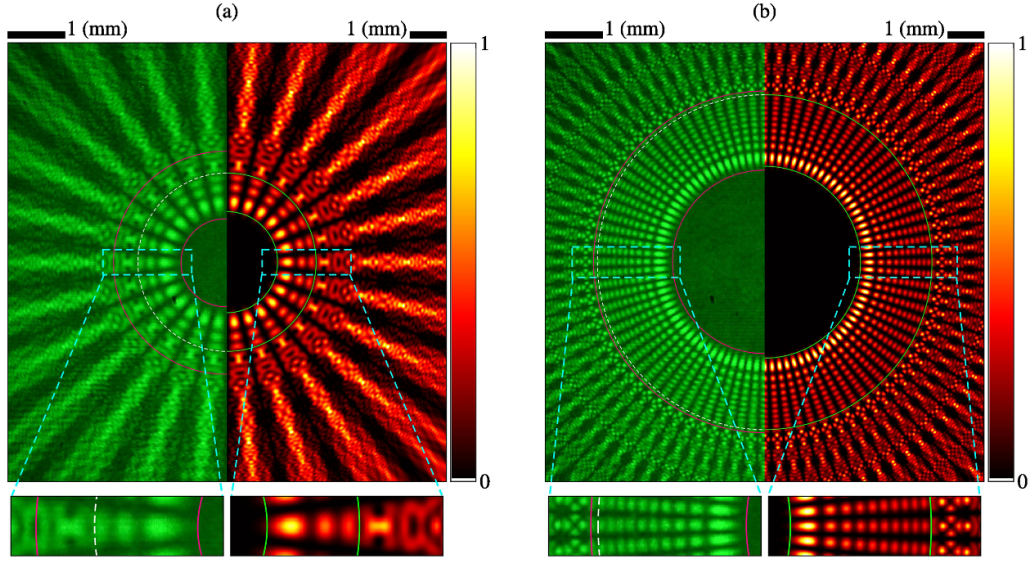


Figure 6. Comparison between experimental (left halves, recorded at $z = 45$ cm) and theoretical (right halves, computed at $z = 1$ m) intensity patterns for RCBs with (a) $m = 13$ and (b) $m = 44$. Insets below each panel highlight the intermediate radial intensity zones used for peak counting. Solid semicircles in both experimental and theoretical patterns indicate the radial boundaries detected by the MATLAB algorithm. In the experimental patterns, the white dashed semicircles represent the visually estimated outer boundary of the intermediate radial region, suggesting that the algorithm overestimates the number of peaks in (a) and fails to detect the onset of peak splitting in (b), resulting in a misidentified outer boundary. These mismatches arise due to experimental limitations such as resolution, noise, camera artifacts, and structural distortions during image acquisition.

in certain regions. As a result, each value of m requires customized tuning of the smoothing parameters and peak detection thresholds. Even then, occasional errors occur in the analysis. Two examples of such errors in the analysis of experimentally recorded beams are shown in figure 6, alongside their theoretical counterparts for $m = 13$ and $m = 44$. The initial and final radial boundaries (r_1 and r_2) of the intermediate intensity zone in the experimental and theoretical images are identified by the MATLAB algorithm and indicated by two solid semicircles on either side of the highlighted region. In figures 6(a) and (b), the right half shows the theoretical beam at $z = 1$ m, while the left half displays the experimental beam recorded at $z = 45$ cm. As seen in figure 6(a) for $m = 13$, the theoretical analysis successfully detects the outer boundary of the intermediate radial spots, whereas the experimental analysis fails to do so. As a result, three intermediate radial spots are detected for the theoretical beam pattern and four for the experimental beam pattern. However, visual inspection clearly shows that both the experimentally generated and the theoretically computed beam patterns contain the same number of intermediate spots. In the experimental pattern, this visually inferred outer boundary is highlighted by a white dashed semicircle, which does not coincide with the boundary detected by the MATLAB algorithm, shown in solid magenta. In the case of $m = 44$ in figure 6(b), careful examination of the insets reveals that the theoretical beam analysis correctly identifies the final boundary of the intermediate intensity zone and excludes the spots that are in the process of splitting. In contrast, the degraded quality of the experimental image, caused by the aforementioned factors, leads to a misidentification and analysis error. In this case as well, the visually estimated boundary is shown by a white dashed semicircle in the

experimental pattern, highlighting a more accurate termination point for the intermediate radial region than the boundary detected by the MATLAB algorithm.

Therefore, despite visual agreement between experimental and theoretical patterns, the graph in figure 5 was constructed based on the MATLAB analysis of the theoretical beam model and visual verification of experimental images for $m \leq 70$. For $m > 70$, as discussed earlier, only calculations based on theoretical beams were used to preserve consistency and accuracy.

In figure 5(a), most data points, except for two small gray shaded regions that will be discussed later, exhibit a clear regular stepwise behavior as the number of grating spokes m increases, each step consists of two consecutive values of m for which the number of intermediate radial intensity spots n remains constant. From each step base, for example at $m = 27$ and $m = 28$, the plot ascends through three consecutive upward steps, each covering two values of m , and then descends by a single step, such as at $m = 35$ and $m = 36$. This repeating pattern of three up, one down creates a special form of staircase behavior that is not uniform but rather depends on the parity (evenness or oddness) of m . This structure is illustrated more clearly in figure 5(c), which shows a magnified portion of the graph corresponding to m values from 27 to 36. This green shaded panel highlights a complete instance of the observed stepwise pattern. The number of intermediate radial intensity spots n remains constant for two consecutive values of m , then increases step by step over three such intervals, and finally decreases once over the next pair. Specifically, the base of this cycle occurs at $m = 27, 28$, where $n = 6$. It then increases to $n = 7$ at $m = 29, 30$, to $n = 8$ at $m = 31, 32$, and to $n = 9$ at $m = 33, 34$, before decreasing back to $n = 8$ at $m = 35, 36$. Notably, this final step, which marks the

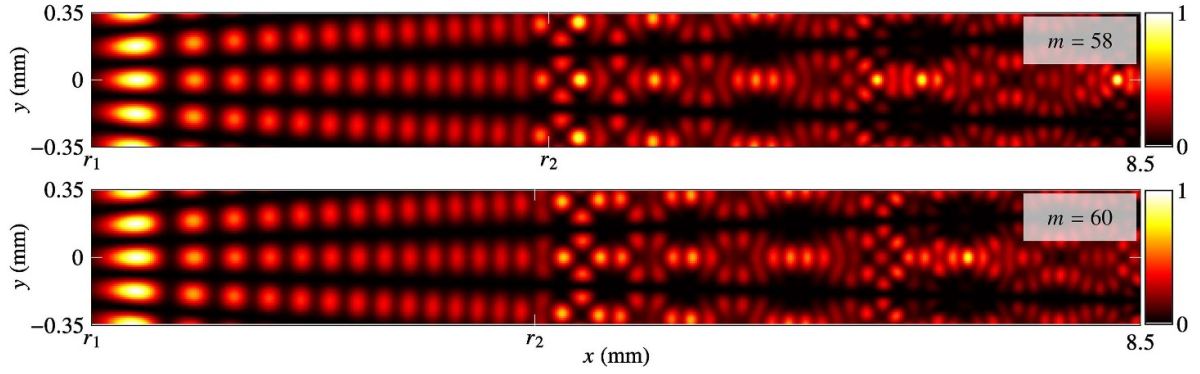


Figure 7. Beam intensity insets for $m = 58$ and $m = 60$, showing a drop in radial peak count (from 15 to 14) and a reduced outer radius r_2 , marking the start of a new skew-step cycle.

drop in n , also serves as the base of the next rising sequence in the following cycle. This local pattern reflects the global three up, one down rule discussed earlier, offering a clear view of the discrete and non-monotonic evolution of n with respect to m .

It is worth noting that while a purely increasing stepwise trend may appear predictable and intuitive, the emergence of periodic downward steps within this progression introduces an unexpected and intriguing feature, one that reflects the internal organization of the beam.

While the overall trend is consistent, certain regions deviate from this structured behavior, as highlighted in the gray shaded areas. The first gray region, which corresponds to values of m from 2 to 21, shows some irregularities. In this range, the complexity of the beam's intensity pattern is relatively low due to the small value of m , which likely results in reduced structural complexity, causing inconsistencies in the observed values of n . Such deviations are not unexpected, as simpler beam structures may lack the robustness required to consistently form well-defined intensity peaks. The pink shaded area, shown in figure 5(b), covering values of m from 3 to 19, provides a closer look at the fluctuations occurring at lower values of m . Here, the behavior of n reveals several inconsistencies when compared to the general trend, which is likely due to the reduced complexity of the beam structure at smaller m , allows these irregularities to emerge. The yellow circles in figure 5(b) represent the expected values if the data followed a more predictable trend, showing the points at which the observed data diverge from the ideal progression. The second gray region, spanning values of m from 66 to 74, indicates deviations that may be attributed to potential computational errors or deviations from the expected trend. To provide further clarity, the blue shaded inset, shown in figure 5(d), highlights that if the data followed the previous trend, the red circles for $m = 66, 73$, and 74 would align with the yellow circles, which represent the expected values. This suggests a break from the otherwise consistent progression. With the proper adjustment of these yellow points in both the pink and blue regions, the overall behavior would be smooth and continuous. This approach is inspired by the mathematical patterns found in other values of m , helping to correct the incorrect behaviors and complete the overall trend. It should also be noted that, in general, for large m , the

counting of n is more reliable because the peaks and valleys defining each spot have significant intensity differences.

Additionally, figure 5(e) presents a separate analysis of the behavior of $n(m)$ for even and odd values of m , depicted with red and blue triangles, respectively. Both plots display a similar periodic pattern, in which four consecutive data points are followed by a discernible drop at the beginning of the next four-point segment. This alternating structure reflects a consistent yet non-linear trend in the variation of radial intensity peaks and suggests a non-trivial relationship between intermediate and MISs in RCBs. The overall pattern can be characterized by a pair of laterally shifted skew-step functions, each corresponding to even and odd values of m .

This trend is further illustrated in figure 7, which shows inset beam intensity profiles for $m = 58$ and $m = 60$. Although the number of grating spokes increases from 58 to 60, the number of intermediate radial intensity peaks up to radius r_2 decreases from 15 to 14. This local reduction clearly marks the start of a new cycle, in agreement with the downward steps observed in panel figure 5(e). Notably, this reduction in the number of peaks is accompanied by a visible decrease in the value of r_2 , compared to beams with preceding values of m . This behavior reinforces the non-monotonic and discrete nature of the stepwise progression described earlier, and can also be observed in Visualization 2 of figure 3 (Multimedia available online).

4.1. Examining the behavior of even and odd values of m

To explore this periodicity in greater detail, we analyze the behavior of even and odd values of m separately. For the even values, the number of intermediate radial intensity peaks follows a clear repeating pattern. The start and end points of each four-point period are listed in table 1.

Upon closer examination, the starting point of each period follows the expression $4(2k - 1)$, where k is a natural number. The slope of each linear segment within these periods is $\frac{1}{2}$, indicating a gradual increase in the number of peaks. The corresponding endpoints can be determined using $4(2k - 1) + 6$. It is important to note that each new segment begins one unit lower than the end of the previous one, introducing a discrete drop at the start of each cycle without implying an overall

Table 1. Initial and final values of m for each cycle of even m .

Cycle number k	Initial m	Final m
4	28	34
5	36	42
6	44	50
7	52	58

decreasing trend. As a result, the relationship for $n(m)$ for even values of m can be expressed as

$$n(m) = \frac{1}{2}m - 2k. \quad (6)$$

For the odd values of m , the behavior can be deduced by examining the preceding even value of m . In this case, the relationship for each four-point period behaves similarly, and it can be expressed as

$$n(m) = \frac{1}{2}(m + 1) - 2k. \quad (7)$$

4.2. General relationship for $n(m)$

Based on this periodic behavior, a general mathematical relationship for $n(m)$ can be established as follows:

$$n(m) = \begin{cases} \frac{1}{2}m - 2k, & \text{If } m \text{ is even and} \\ & 4(2k - 1) \leq m \leq 4(2k - 1) + 6 \\ \frac{1}{2}(m + 1) - 2k, & \text{If } m \text{ is odd and} \\ & 4(2k - 1) - 1 \leq m \leq 4(2k - 1) + 5 \end{cases} \quad (8)$$

where $k = 4, 5, 6, \dots$. The reason for excluding $k \leq 3$ lies in the irregularities observed in the first gray region, which indicates that these values do not follow the established trend due to the lower complexity of the beam structure.

This detailed analysis highlights the systematic trends and deviations in the intermediate number of radial intensity peaks, providing insights into the structural dynamics of the RCBs. The observed stepped and periodic behavior reflects the intricate interplay between the beam parameters and the resulting diffraction patterns.

It is also worth noting that a similar problem has been studied in the context of counting the number of vortices in fractional vortex beams (FVBs) as reported in [45]. Interestingly, in that work, the total number of vortices also exhibits a step-like behavior, with jumps occurring when the topological charge is very close to (before and after) even integer values. However, unlike the behavior reported here, their stepwise trend does not include the kind of sudden downward transitions observed in our study. This comparison highlights that the emergence of step-like patterns in different structured light properties, whether in the number of vortices in FVBs as in [45] or in the number of intermediate radial intensity spots in our work, is a fascinating phenomenon in the optical part of nature.

5. Conclusion

This investigation concludes by identifying the skew-step function that governs the number of intermediate radial intensity spots in RCBs as a function of the number of spokes in the generating radial phase grating. Our findings indicate that the number of these spots increases over three consecutive steps and then decreases in the fourth, forming a repeating three up, one down pattern. This discrete and non-monotonic trend is elegantly captured by a pair of skew-step functions, each associated with either even or odd values of m , which share the same form but are laterally shifted.

This underscores the systematic and well-defined behavior of RCBs, highlighting their unique properties, which emerge from the diffraction of plane waves by binary phase radial gratings. Interestingly, this staircase-like regularity arises within a structured beam itself, suggesting that light, as a part of nature, possesses an inherent mathematical order.

The results of this research reflect the broader natural tendency toward order and symmetry, a recurring theme across physical phenomena. Just as plant growth and planetary arrangements follow inherent mathematical laws, RCBs also demonstrate a disciplined, structured behavior. By revealing this underlying order, our investigation not only enhances our understanding of structured light but also bridges the gap between natural principles and optical beam dynamics. This connection emphasizes the intrinsic unity between the physical laws governing nature and those governing light propagation, opening new avenues for future exploration of structured light and its applications. In addition to its conceptual significance, the discovered relationship between n and m enables precise optical pattern engineering. This allows pre-determination of the grating spokes to generate a desired two-dimensional symmetric polar array of intensity spots, forming an optical lattice [16] with m azimuthal rows and n radial spots, for applications such as multifocus beam generation and optical trapping of particles, even including atomic-scale objects [46].

Moreover, the counter-intuitive nature of this behavior makes it a valuable demonstration tool for educational purposes, helping students appreciate that scientific phenomena often reveal unexpected patterns.

Data availability statement

The data supporting the findings of this study were generated using MATLAB during the calculation process and can be reproduced using the MATLAB codes available in the online repository [47]. A supplementary PDF explaining the codes and their structure is also provided in the repository to facilitate reproduction and further use.

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Conflict of interest

The authors have no conflicts to disclose.

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